

From Algorithmic Scaffolding to Embodied Authority: Negotiating Human Expertise in AI-Mediated Fitness Coaching

Wenyu Liao

School of Communication
Northwestern University
Evanston, Illinois, USA
wenyuliao2030@u.northwestern.edu

Na Liu

The Department of Work and Organisation Studies
KU Leuven
Leuven, Belgium
Na.liu@kuleuven.be

Abstract

Generative Artificial Intelligence (GenAI) is increasingly integrated into professional work, yet its impact is often framed through task substitution and automation. Drawing on interviews, multi-sited observation, and professional immersion with fitness coaches, this poster examines how GenAI is incorporated into embodied service labor in practice. We find that trainers use GenAI as algorithmic scaffolding—delegating codified tasks such as program drafting and nutritional calculation—while retaining authority through embodied judgment, relational interaction, and real-time adaptation. When AI-generated plans encounter living bodies, standardized prescriptions frequently break down, revealing limits tied to bodily sensation, affect, and situational context. Rather than rejecting AI, practitioners actively negotiate these boundaries, repositioning their expertise around embodied articulation and client-facing work. These findings suggest that AI does not replace professional labor, but reorganizes where authority resides—standardizing baseline competence while amplifying the importance of embodied expertise. The study contributes early empirical insight into how embodiment shapes the limits of AI-mediated codification and human–AI collaboration in professional work.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; • **Computing methodologies** → Artificial intelligence.

Keywords

Generative AI, Progressive encapsulation, Embodied expertise, Human-AI collaboration, Professional authority

ACM Reference Format:

Wenyu Liao and Na Liu. 2026. From Algorithmic Scaffolding to Embodied Authority: Negotiating Human Expertise in AI-Mediated Fitness Coaching. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798843>

1 Introduction

Generative Artificial Intelligence (GenAI) has become increasingly embedded in everyday work practices. Rather than functioning

solely as assistive tools that execute discrete tasks delegated by humans, GenAI systems are becoming deeply internalized within organizational routines [4, 9, 25]. As a result, they not only support human labor but also shape how workers think, make decisions, and carry out their activities. Across domains such as consulting, design, healthcare, and education, GenAI is now integrated into creative content production, customer service interactions, progress tracking, and the provision of learning materials [7, 8, 10].

How should we understand this transformation of work in the presence of GenAI? Existing scholarship has largely approached this question through a task-centered paradigm, asking whether and how specific activities can be automated or substituted by AI systems [1, 16]. Building on this premise, more recent work argues that GenAI *progressively encapsulates* professional activities—absorbing routines, judgment criteria, and even evaluative authority into algorithmic systems’ “black box” [13]. This shift raises an open and unsettled question: what aspects of professional labor remain resistant to such encapsulation?

We examine fitness coaching as an empirical site where GenAI is increasingly incorporated into everyday practice. Fitness coaching has seen growing use of GenAI for workout and nutrition planning, progress tracking, and motivational support [3, 19]. At the same time, coaching is fundamentally embodied and relational [2, 12]: practitioners rely on bodily observation, tactile feedback, real-time adjustment, and affective engagement to guide clients. This combination makes fitness coaching a productive case for examining how human labor and GenAI systems interact, coexist and reshape one another beyond narratives of task replacement.

This poster presents emerging qualitative findings from interviews and field observations with fitness coaches to explore how GenAI is integrated into embodied professional work. Rather than treating AI as replacing or simply augmenting human labor, we focus on how practitioners delegate codified knowledge to AI while retaining authority over embodied judgment and client-facing interaction. In doing so, we aim to surface early insights into how professional expertise, labor value, and human-AI relations are being reconfigured in AI-mediated work.

2 Related work

2.1 Progressive Encapsulation, Task-Centered Accounts

Recent work conceptualizes Generative AI (GenAI) not simply as a tool for automating discrete tasks, but as part of a broader process of *progressive encapsulation*, through which professional routines, evaluative criteria, and operational knowledge are increasingly



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3798843>

absorbed into algorithmic systems and platform infrastructures [13]. Progressive encapsulation describes how professional routines and evaluative criteria become embedded within algorithmic systems, gradually standardizing practice and obscuring execution processes [15] [24].

Building on this perspective, task-centered studies map AI capabilities onto occupational task lists to estimate substitution risk or productivity gains [9, 11]. While this work usefully demonstrates the breadth of occupational exposure to generative technologies, it also shows that exposure alone does not guarantee adoption or transformation. Whether AI reshapes work in practice depends on workflow integration, professional judgment, institutional incentives, and the situated contingencies of use [4].

2.2 The Limits of Codification and Embodiment in Professions

Progressive encapsulation shifts attention from what AI can do to how codified routines become durable and authoritative in practice—and who controls that process. Yet even this framework has paid limited attention to the human body as a potential boundary of encapsulation. Much of the AI-and-work literature abstracts labor into informational or procedural elements, overlooking how work is enacted through embodied, sensory, and relational engagement [5, 22].

Research on embodiment and emotional labor suggests that professional expertise often relies on forms of knowledge that are difficult to formalize or audit. Classic accounts distinguish between surface and deep acting in affective labor [14], while studies of embodied expertise emphasize tacit, nonverbal, and context-specific know-how. For example, Menon and Sariego [17] show how cosmetic surgeons strategically deploy embodied and emotional labor to sustain trust, authority, and professional status—demonstrating that such practices are not peripheral but central to value creation.

From the perspective of codification, these embodied and relational forms of competence may generate friction for encapsulation. While procedural knowledge can often be standardized and scaled, practices that depend on co-presence, sensory feedback, and longitudinal relationships resist straightforward translation into rules or templates. As a result, codifiability is better understood as graded rather than binary: some elements of professional work are readily encapsulated, while others remain anchored in embodied interaction.

This insight motivates our empirical focus on fitness coaching, a domain where GenAI is increasingly used to generate plans, recommendations, and motivational content, yet where effective practice continues to depend on bodily observation, tactile correction, and affective engagement. By examining how coaches integrate, negotiate, and sometimes resist AI-generated prescriptions in practice, we use fitness coaching as a lens to explore how embodiment shapes the limits of progressive encapsulation in AI-mediated labor.

3 Methods

3.1 Data Collection

This study draws on a multi-method qualitative dataset combining semi-structured interviews, multi-sited participant observation,

and researcher immersion. We conducted 19 semi-structured interviews in Mandarin Chinese with personal trainers across three market segments: mass-market commercial gyms, specialized health-management clinics, and elite bodybuilding or performance studios. Participants were recruited via a public study announcement on social media, with the explicit criterion that they had at least some experience using generative AI in their work. Interviews were conducted via Zoom, audio- and video-recorded with informed consent, and lasted approximately 60–90 minutes. The sample captures variation in professional orientation, years of experience, and attitudes toward AI (Table 1). Our sample focused on trainers with some experience using GenAI, which allowed us to examine how AI becomes embedded in practice. However, this skews the study toward adopters and limits insight into the reasoning of outright non-users. Future research should examine resistant practitioners to further clarify the technical and embodied limits of AI-mediated coaching.

To situate interview accounts within everyday practice, we also conducted approximately 100 hours of participant observation across two sites: a high-traffic urban gym in Philadelphia, USA (60 hours), and a community-centered training facility in Jiangsu, China (40 hours). Observations focused on trainers' embodied interactions with clients, including physical correction, tactile cueing, real-time adjustment, and motivational engagement. Detailed fieldnotes documented how standardized plans were enacted, modified, or overridden in practice.

The first author additionally engaged in sustained professional immersion, including a nine-month intensive study of National Strength and Conditioning Association (NSCA) materials culminating in certification, as well as participation in an NSCA conference. This immersion supported analytic sensitivity to professional norms, evaluative standards, and AI-mediated informational infrastructures from both practitioner and user perspectives.

3.2 Analytic Approach

Analysis followed an iterative, interpretive approach informed by constructivist grounded theory [6] and prior work on AI-mediated labor, progressive encapsulation and embodiment. Interview transcripts, fieldnotes, and artifacts were coded inductively to capture how trainers integrated AI-generated outputs into their everyday work and how they exercised judgment in client-facing situations.

Across analytic iterations, we compared cases and practices to trace recurring patterns in how AI-generated plans were adopted, modified, or resisted over time. This process-oriented analysis foregrounded moments where standardized prescriptions encountered embodied constraints, prompting practitioners to reassert authority through situated judgment and relational work. Analytic memos and comparative matrices were used as heuristic tools to surface contrasts between codifiable and embodied dimensions of coaching practice.

4 Findings

Our findings show that the integration of Generative AI (GenAI) into fitness coaching does not follow a logic of task substitution or linear automation. Instead, trainers engage in an ongoing process

Table 1: Interviewee Overview

Alias	Primary Role	Experience	AI Orientation
Lemon	Professional coach (Bodybuilding)	10+ yrs	Humanistic Skeptic
Chunshu	Independent coach (Body Shaping)	1–3 yrs	Proactive Integrator
Tian	PhD student (PT) & Part-time coach	1–3 yrs	Critical Pragmatist
Jing	Fitness equipment professional	10+ yrs	Proactive Integrator
Qingyan	Independent coach	1–3 yrs	Proactive Integrator
Coach Zou	Senior private coach	10+ yrs	Humanistic Skeptic
Ivan	Private coach & entrepreneur	10+ yrs	Proactive Integrator
Henry	PhD student & part-time coach	1–3 yrs	Humanistic Skeptic
KK	Independent coach	1–3 yrs	Critical Pragmatist
Coco	Professional coach (former AI PM)	3–5 yrs	Critical Pragmatist
Coach Wei	Graduate student (Sports Training)	3–5 yrs	Critical Pragmatist
CC	Senior coach (Martial Arts)	10+ yrs	Humanistic Skeptic
Meggie	Private coach	3–5 yrs	Critical Pragmatist
Walker	Independent coach	5–10 yrs	Critical Pragmatist
Zhou	Rehabilitation therapist	5–10 yrs	Critical Pragmatist
Guali	Avid rock climber / Data student	1–3 yrs	Proactive Integrator
Yang	Coach trainer	10+ yrs	Proactive Integrator
Liqing	Rehabilitation specialist	10+ yrs	Critical Pragmatist
Yuhua	Independent part-time coach	1–3 yrs	Critical Pragmatist

of relational negotiation, in which AI-generated outputs and embodied professional expertise are continually aligned, adjusted, and reasserted in practice. Rather than treating GenAI as a standalone decision-maker or replacement tool, trainers strategically position it within the labor process—delegating codified knowledge to AI while retaining authority through embodied judgment, interaction, and client-facing work.

4.1 Algorithmic Scaffolding: Delegating Codified Knowledge

Trainers commonly used GenAI to generate the basic structure of training and nutrition plans. Rather than replacing expertise, AI operated as an *algorithmic scaffold*: a way to rapidly produce standardized, information-dense drafts that trainers then reviewed, edited, and contextualized. Participants framed this use as a pragmatic response to time pressure and rising efficiency expectations, not as a transfer of professional responsibility.

Across interviews, practitioners described a stable “copy-and-edit” workflow. AI drafted baseline programs—workout splits, periodization templates, dietary calculations—while trainers retained responsibility for judging appropriateness and making final adjustments. As one coach put it, “It’s like copying someone’s homework in high school... I just copy it once” (Walker), underscoring efficiency rather than authorship. Another described AI output as a “70–80 point first draft,” “usable, but not final,” requiring revision to accommodate movement quality, injury history, and situational constraints (Yuhua). Trainers thus treated AI plans as provisional artifacts, not authoritative prescriptions.

Trainers stressed that AI provided structure, not practice. As Spring noted, tools such as DeepSeek were helpful for generating “diet plans, training plans, and underlying training theories,” but still required human judgment to decide “what actually works

for this person.” AI-generated plans were repeatedly cast as the “skeleton” of a program, with professional expertise supplying the “flesh and blood” through contextual knowledge and embodied assessment. AI was especially valued for “tedious calculation work”—macronutrient distributions, caloric timing, basic load progressions—freeing trainers to devote more attention to client interaction and real-time adjustment during sessions.

At the same time, widespread access to high-quality AI plans was seen as reshaping the value of foundational expertise. Several trainers argued that AI now sets a scalable floor for technical competence, diminishing the distinctiveness of standardized knowledge. One participant estimated that AI could replace “80–90%” of low-customization, template-based coaching services (Walker), a concern voiced most strongly in commercial gyms where standardized programming has long been central.

As foundational knowledge became standardized, trainers increasingly positioned themselves as auditors and contextualizers of AI output, shifting effort from producing information to evaluating and adapting it for embodied delivery.

4.2 Embodied Friction: Where Standardization Breaks Down

When AI-generated plans encountered living bodies, their limits became visible. Trainers consistently framed these moments of breakdown not as technical glitches, but as AI’s inability to access bodily sensation, situational context, and lived experience. Here, the authority of standardized intelligence weakened and embodied expertise reasserted itself.

As one trainer, Maggie, put it, “AI has no feelings; it cannot tell you how a muscle feels in your body, how your muscles change in your body when you’re taking action.” Physical assessment relied on multi-sensory perception—watching movement from multiple

angles, listening for joint sounds, and feeling force production through touch—forms of embodied judgment unavailable to algorithmic systems (Yuhua). Trainers noted that AI struggled to detect compensatory patterns or set realistic starting points for clients with pain, asymmetry, or injury histories, often producing plans that were technically coherent but practically unsafe or ineffective.

Rigid numerical prescriptions intensified this friction. Trainers criticized AI recommendations that demanded precise adherence to quantified targets—exact grams of protein, milliliters of water—describing them as “unrealistic” and “not humanized.” As Coach Zou remarked, “Eating x grams of protein, are people really going to weigh and calculate that?” Others emphasized that bodily data often diverged from lived reality due to psychological state, stress, or relational dynamics, rendering technically “perfect” plans impractical or counterproductive in everyday life.

Beyond biomechanics, trainers highlighted the relational and affective dimensions of embodied work. Guiding clients through shared physical struggle—fatigue, discomfort, perceived risk—was described as central to trust-building and long-term transformation. Guali, for instance, characterized fatigue as clients’ “cells fighting bravely,” framing sessions as a shared physical ordeal that builds intimacy and trust, particularly in high-risk sports such as rock climbing. This “shared risk,” especially salient in high-intensity or technically demanding activities, generated forms of intimacy and motivation that AI-generated encouragement could not replicate. Trainers thus framed their role as helping clients feel progress rather than merely understand it.

They also identified a relational gap in AI-mediated interaction. AI-generated encouragement was often experienced as polite but emotionally hollow. Meaningful support, trainers argued, required the capacity to challenge, confront, or disagree with clients—what one coach termed “human collision” (CC). Face-to-face interaction produced a shared energy and emotional intensity that trainers viewed as essential for sustained engagement, yet fundamentally inaccessible to AI systems.

Across both technically oriented and relationally oriented coaching, friction marked the point at which practitioners reasserted authority by translating, adapting, or rejecting standardized outputs. While AI could encapsulate baseline intelligence, it could not internalize the proprioceptive, affective, and relational labor required for deep transformation. The practitioner’s contribution lay precisely in this embodied articulation—helping clients interpret bodily signals, recalibrate expectations, and internalize change—producing not a generic digital output but a bespoke, lived experience resistant to full automation.

5 Discussion

This study offers an empirical lens on professional labor under conditions of progressive encapsulation, showing that GenAI transforms work not by fully automating expertise, but by redrawing the boundary between codifiable routines and the embodied, relational practices that resist encapsulation. While dominant accounts of AI and work rely on task-centered frameworks to assess automation risk [9, 20], our findings point to a different constraint. Encapsulation advances most effectively where work can be standardized and codified, but encounters limits when algorithmic prescriptions meet

the lived, embodied realities of human bodies. At this boundary, professional labor is neither replaced nor rendered obsolete. Instead, authority is redistributed and renegotiated through embodied judgment, mediation, and relational work.

This dynamic helps explain a central paradox of GenAI in professional practice. On the one hand, algorithmic scaffolding can improve short-term performance by standardizing technical competence and reducing cognitive effort [4]. On the other hand, reliance on standardized outputs risks weakening the situated expertise required to adapt when algorithmic support breaks down or proves insufficient [15, 23]. These tensions suggest that the impact of GenAI cannot be captured by a simple binary of substitution versus augmentation. Rather, GenAI reshapes professional labor by shifting where judgment, responsibility, and authority are exercised.

Our findings further highlight the limits of GenAI in domains where work is fundamentally embodied and relational. While GenAI excels at synthesizing informational knowledge drawn from protocols, datasets, and best practices, it remains blind to bodily sensation, affective response, and the sensory contingencies of work in motion [18, 21]. Even technically correct prescriptions often fail when they encounter fatigue, pain, fear, or fluctuating motivation in practice. Indeed, embodied expertise involves tacit, multi-sensory knowledge that resists full codification [12, 22].

Our findings suggest implications for the design of AI systems in embodied domains. Rather than optimizing for autonomous prescription, tools should support auditable scaffolding—making assumptions, uncertainty, and risk boundaries visible so practitioners can efficiently evaluate and adapt outputs. In high-variability contexts such as sports and physical training, systems might incorporate uncertainty signaling or prompts for in-person reassessment when recommendations intersect with injury risk, fatigue, or psychological fluctuation. Designing for hybrid authority—where humans remain accountable interpreters rather than invisible overseers—may better align AI systems with embodied professional practice.

Taken together, these findings suggest that the limits of AI in professional labor are not merely technical, but arise from the embodied and relational conditions through which expertise is enacted. Rather than viewing AI as replacing human work, our study points to how AI reorganizes labor by standardizing baseline competence while amplifying the importance of embodiment at the boundaries of encapsulation. For HCI and social computing research, this highlights the need to attend not only to what AI systems can automate, but also to how human bodies, relationships, and situated practice shape the future of AI–human collaboration.

References

- [1] Daron Acemoglu and Pascual Restrepo. 2019. Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives* 33, 2: 3–30. <https://doi.org/10.1257/jep.33.2.3>
- [2] Ellen Penn Berry. 2018. “But you still have the same insecurities”—Navigating Personal Trainer and Client Body Image through Emotional and Aesthetic Labor. Retrieved February 27, 2026 from <https://commons.lib.jmu.edu/master201019/570/>
- [3] Attila Biró. 2024. Gamification, genai and reinforcement learning as motivational assets in performance sports. In *2024 IEEE 18th International Symposium on Applied Computational Intelligence and Informatics (SACI)*, 000315–000320. Retrieved February 27, 2026 from [https://ieeexplore.ieee.org/abstract/document/10619802/?casa_token=\\$9KjkDal2XdsAAAAA:N2xqRWsFe96cy8zETq3ry5EadTVEm0jwWMd1Zhz5C5_4-](https://ieeexplore.ieee.org/abstract/document/10619802/?casa_token=$9KjkDal2XdsAAAAA:N2xqRWsFe96cy8zETq3ry5EadTVEm0jwWMd1Zhz5C5_4-)

- jwkkWce39Ge6l1bAdV7tN0c_ELXw
- [4] Erik Brynjolfsson and Zoe Hitzig. 2025. AI's Use of Knowledge in Society. *The Economics of Transformative AI. National Bureau of Economic Research, Cambridge, MA*. Retrieved January 23, 2026 from <https://www.nber.org/system/files/chapters/c15303/revisions/c15303.rev0.pdf>
 - [5] Erik Brynjolfsson and Andrew McAfee. 2014. *The second machine age: Work, progress, and prosperity in a time of brilliant technologies*. WW Norton & company. Retrieved January 23, 2026 from [https://books.google.com/books?hl=\\$en&lr=\\$&id=\\$WiKwAgAAQBAJ&oi=\\$fnd&pg\\$=\\$PA1&dq\\$=\\$Brynjolfsson+%26+McAfee,+2014&ots=\\$4-ZwSg5v8e&sig=\\$9qrsia2N7JyRf51NCgIrgVRE-AY](https://books.google.com/books?hl=$en&lr=$&id=$WiKwAgAAQBAJ&oi=$fnd&pg$=$PA1&dq$=$Brynjolfsson+%26+McAfee,+2014&ots=$4-ZwSg5v8e&sig=$9qrsia2N7JyRf51NCgIrgVRE-AY)
 - [6] Kathy Charmaz. 2006. *Constructing grounded theory: A practical guide through qualitative analysis*. sage. Retrieved January 23, 2026 from [https://books.google.com/books?hl=\\$en&lr=\\$&id=\\$2ThdBAQAQBAJ&oi=\\$fnd&pg\\$=\\$PP1&dq\\$=\\$grounded+theory+\(Charmaz,+2006\)+&ots=\\$f-o_5KsBCY&sig=\\$NzDMgQeO7nWIL9vkUyrC-bv1QL4](https://books.google.com/books?hl=$en&lr=$&id=$2ThdBAQAQBAJ&oi=$fnd&pg$=$PP1&dq$=$grounded+theory+(Charmaz,+2006)+&ots=$f-o_5KsBCY&sig=$NzDMgQeO7nWIL9vkUyrC-bv1QL4)
 - [7] Yu Cui, Yaru Meng, and Lingjie Tang. 2025. Reconsidering teacher assessment literacy in GenAI-enhanced environments: A scoping review. *Teaching and Teacher Education* 165: 105163.
 - [8] Peter Dalsgaard. 2025. GenAI and the Crisis of Creative Labor: Automation, Augmentation, and the Artist's Role. In *Adjunct Proceedings of the Sixth Decennial Aarhus Conference: Computing X Crisis*, 1–5. <https://doi.org/10.1145/3737609.3747116>
 - [9] Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. 2023. Gpts are gpts: An early look at the labor market impact potential of large language models. *arXiv preprint arXiv:2303.10130* 10. Retrieved January 23, 2026 from <https://files.lafm.com.co/assets/public/2023-03/2303.10130.pdf>
 - [10] Guiseppe Getto, Susan Kelley, and Bremen Vance. 2025. How to Write With GenAI: A Framework for Using Generative AI to Automate Writing Tasks in Technical Communication. *Journal of Technical Writing and Communication*: 00472816251332208. <https://doi.org/10.1177/00472816251332208>
 - [11] Kazem Haki, Dorsa Safaei, Adolfo Magan, and Martin Griffiths. 2025. Integrating Generative AI Into Enterprise Platforms: Insights From Salesforce. *Information Systems Journal* 35, 5: 1497–1512. <https://doi.org/10.1111/isj.12593>
 - [12] Geraint Harvey, Sheena J. Vachhani, and Karen Williams. 2014. Working out: aesthetic labour, affect and the fitness industry personal trainer. *Leisure Studies* 33, 5: 454–470. <https://doi.org/10.1080/02614367.2013.770548>
 - [13] Pamela Hinds and Georg Von Krogh. 2024. Generative AI, Emerging Technology, and Organizing: Towards a theory of progressive encapsulation. *Organization Theory* 5, 4: 26317877241293478. <https://doi.org/10.1177/26317877241293478>
 - [14] Ann Hochschild, Nina Irwin, and Mark Ptashne. 1983. Repressor structure and the mechanism of positive control. *Cell* 32, 2: 319–325.
 - [15] Oscar Lederman, Alessandro Llana, James Murray, Robert Stanton, Ritesh Chugh, Darren Haywood, Amanda Burdett, Geoff Warman, Joanne Walker, and Nicolas H. Hart. 2025. Promises and perils of generative artificial intelligence: a narrative review informing its ethical and practical applications in clinical exercise physiology. *BMC Sports Science, Medicine and Rehabilitation* 17, 1: 131. <https://doi.org/10.1186/s13102-025-01182-7>
 - [16] Elina Mäkelä and Fabian Stephany. 2025. Complement or substitute? How AI increases the demand for human skills. <https://doi.org/10.48550/arXiv.2412.19754>
 - [17] Alka V. Menon and Chloe Sario. 2022. Engendering connection: The embodied emotional labor of US cosmetic surgeons. *Social Science & Medicine* 306: 115092.
 - [18] Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, and Kristian Lum. 2021. Algorithmic fairness: Choices, assumptions, and definitions. *Annual review of statistics and its application* 8, 1: 141–163.
 - [19] Luca Puce, Nicola Luigi Bragazzi, Antonio Currà, and Carlo Trompetto. 2025. Harnessing generative artificial intelligence for exercise and training prescription: applications and implications in sports and physical Activity—A systematic literature review. *Applied Sciences* 15, 7: 3497.
 - [20] Sebastian Raisch and Sebastian Krakowski. 2021. Artificial Intelligence and Management: The Automation–Augmentation Paradox. *Academy of Management Review* 46, 1: 192–210. <https://doi.org/10.5465/amr.2018.0072>
 - [21] Laavanya Ramaul, Paavo Ritala, Angelos Kostis, and Päivi Aaltonen. 2025. Rethinking How We Theorize AI in Organization and Management: A Problematic Review of Rationality and Anthropomorphism. *Journal of Management Studies*: joms.13246. <https://doi.org/10.1111/joms.13246>
 - [22] Lawrence Susskind, Danya Rumore, Carri Hulet, and Patrick Field. 2015. *Managing climate risks in coastal communities: Strategies for engagement, readiness and adaptation*. Anthem Press. Retrieved January 23, 2026 from [https://books.google.com/books?hl=\\$en&lr=\\$&id=\\$-8tRDwAAQBAJ&oi=\\$fnd&pg\\$=\\$PT10&dq\\$=\\$Susskind+%26+Susskind,+2015&ots=\\$JtcpOC9EqL&sig=\\$N6Eg6ll_08AXHQcCk40KGaE](https://books.google.com/books?hl=$en&lr=$&id=$-8tRDwAAQBAJ&oi=$fnd&pg$=$PT10&dq$=$Susskind+%26+Susskind,+2015&ots=$JtcpOC9EqL&sig=$N6Eg6ll_08AXHQcCk40KGaE)
 - [23] Emma Benz Wiles. 2024. Artificial Intelligence in Labor Market Matching. Massachusetts Institute of Technology. Retrieved January 23, 2026 from <https://dspace.mit.edu/handle/1721.1/155874>
 - [24] 2024. GenAI and the New Customer Service Operating Model. *BCG Global*. Retrieved February 27, 2026 from <https://www.bcg.com/publications/2024/transforming-customer-service-operations-with-genai>
 - [25] AI: Work partnerships between people, agents, and robots | McKinsey. Retrieved February 27, 2026 from [https://www.mckinsey.com/mgi/our-research/agents-robots-and-us-skill-partnerships-in-the-age-of-ai?utm_source=\\$chatgpt.com/](https://www.mckinsey.com/mgi/our-research/agents-robots-and-us-skill-partnerships-in-the-age-of-ai?utm_source=$chatgpt.com/)